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Thermodynamic reassessment of Ni-Pr binary system

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ABSTRACT

Based on the available experimental data of phase equilibria and thermodynamic properties from the literature, the Ni-Pr binary system has been thermodynamically assessed using the CALPHAD method. The solution phases, Liquid, FCC_A1, DHCP and BCC_A2 were modeled as substitutional solution phases, for which the excess Gibbs energies were formulated with Redlich–Kister polynomials. All intermetallic phases were described as stoichiometric compounds. Subsequently, a set of self-consistent thermodynamic parameters describing various phases in this binary system has been obtained. The calculated results reproduce well the corresponding experimental data.

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1. Introduction

The rare-earth/transition metal intermetallics have been widely studied from the discovery that several of its members can be used as starting materials for permanent magnets of outstanding quality [1,2]. Furthermore, it has been found that several compounds have quite exceptional properties with respect to reversible absorption of hydrogen gas at room temperature and nearly atmospheric pressures [3,4]. Moreover, many ternary Al–TM–RE systems (TM, transition metal; RE, rare-earth) form amorphous alloys with interesting mechanical properties [5,6] and some rare-earth/transition metal oxides are candidate materials for solid oxide fuel cells [7]. Also, the rare-earth/3d transition metal intermetallic compounds were suggested as being promising candidates for room temperature magnetic refrigeration based on magneto-caloric effect (MCE) [8–10] who has attracted more attention for improved energy efficiency and environmental friendliness compared with traditional compression/expansion gas refrigeration [11,12].

On the other hand, the knowledge of thermodynamic data and phase diagrams is essential for developing and checking models of phase transformations as well as coupling thermodynamics with kinetics [13]. The purpose of the present work is (1) to evaluate recent experimental phase diagram and his relative thermodynamic data and (2) to provide a set of self-consistent parameters for calculation of the phase equilibria and thermodynamic properties of the Ni-Pr binary system using the CALPHAD (CALculation of PHAse Diagram) method [14] and the Thermo-Calc software package [15].

2. Thermodynamic models

2.1. Pure elements

The stable forms of the pure elements at 298.15 K and 1 bar were chosen as the reference states of the system. For the thermodynamic functions of the pure elements in their stable and metastable states, the phase stability equations compiled by Dinsdale [16] and adopted by the SGTE (Scientific Group Thermodata Europe) were used in the present work:

$$\begin{aligned} {}^0G_i^{\phi}(T) &= G_i^{\phi}(T) - H_i^{\mathrm{SER}} \\ &= a + bT + cT\ln T + dT^2 + eT^3 + fT^{-1} + gT^7 + hT^{-9} \end{aligned} \quad (1)$$

where $H_i^{\mathrm{SER}}(298.15\ \mathrm{K})$ is the molar enthalpy of the so-called "Standard Element Reference" (SER), i.e. the enthalpies of the pure elements in their defined reference state at 298.15 K and 1 bar; T is the absolute temperature; $G_i^{\phi}(T)$ is the absolute molar Gibbs energy of the element i (i = Pr and Ni) with structure ϕ in a non magnetic states.

In thermodynamic study, absolute energy is not of importance, so the relative value of Gibbs energy ${}^0G_i^{\phi}(T)$ is adopted in CALPHAD approach. The Gibbs energy of the element i, ${}^0G_i^{\phi}(T)$, in its SER state is denoted by GHSER$_i$, i.e.:

$$\mathrm{GHSER_{Ni}} = {}^0G_{\mathrm{Ni}}^{\mathrm{fcc}}(T) = G_{\mathrm{Ni}}^{\mathrm{fcc}}(T) - H_{\mathrm{Ni}}^{\mathrm{SER}}(298.15\ \mathrm{K}) \quad (2)$$

$$\mathrm{GHSER_{Pr}} = {}^0G_{\mathrm{Pr}}^{\mathrm{dhcp}}(T) = G_{\mathrm{Pr}}^{\mathrm{dhcp}}(T) - H_{\mathrm{Pr}}^{\mathrm{SER}}(298.15\ \mathrm{K}) \quad (3)$$

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Table 1
Curie temperatures of some intermediate phases in the Ni-Pr system.

Compound	NiPr	Ni_7Pr_2	Ni_5Pr	Ni_3Pr
Curie temperature	22 K [18]	85 K [19]	0.44 mK [20]	20 K [21]

2.2. Solution phases

The substitutional solution model was employed to describe the solution phases including Liquid, FCC_A1, BCC_A2 and DHCP. The molar Gibbs energy of the solution phase ϕ (ϕ = Liquid, FCC_A1, BCC_A2 and DHCP) can be expressed as:

$$G_m^\phi = x_{Ni}{}^0G_{Ni}^\phi + x_{Pr}{}^0G_{Pr}^\phi + RT(x_{Ni}\ln x_{Ni} + x_{Pr}\ln x_{Pr}) + {}^EG_m^\phi + {}^{mg}G_m^\phi$$

where G_m^ϕ is the molar Gibbs energy of a solution phase ϕ; ${}^0G_i^\phi$ is the molar Gibbs energy of the element i (i = Pr, Ni) with the structure ϕ in a non-magnetic state; x_i the mole fraction of component i, R gas constant, T temperature; ${}^EG_m^\phi$ the excess Gibbs energy, and ${}^{mg}G_m^\phi$ is the magnetic contribution to the Gibbs energy. The excess Gibbs energy of the phase ϕ can be expressed by the Redlich–Kister polynomials [17] as:

$${}^EG_m^\phi = x_{Ni}x_{Pr}\sum_j {}^jL_{Ni,Pr}^\phi(x_{Ni} - x_{Pr})^j \quad (4)$$

here ${}^jL_{Ni,Pr}^\phi$ (j = 0, 1, 2, ...) is the interaction parameter between elements Ni and Pr and is formulated as temperature dependent:

$${}^jL_{Ni,Pr}^\phi = a_j + b_jT + c_jT\ln T + d_jT^2 + e_jT^3 + f_jT^{-1} + g_jT^7 + h_jT^{-9} \quad (5)$$

where a_j, b_j, c_j, d_j, e_j, f_j, g_j and h_j are model parameters to be optimized. In most cases, only the two first terms of the above equation are used.

2.3. Intermetallic compounds

All intermetallic compounds, Ni_5Pr, Ni_7Pr_2, Ni_3Pr, Ni_2Pr, NiPr, Ni_3Pr_7 and $NiPr_3$ were treated as stoichiometric phases in the Ni-Pr binary system because no available experimental data were reported about the homogeneity range for these compounds. The Gibbs energy of a Ni_APr_B compound is given as:

$${}^0G_m^{Ni_APr_B} = \frac{A}{A+B}{}^0G_{Ni}^{fcc} + \frac{B}{A+B}{}^0G_{Pr}^{dhcp} + a + bT + {}^{mg}G_m^{Ni_APr_B} \quad (6)$$

where ${}^0G_{Pr}^{dhcp}$ and ${}^0G_{Ni}^{fcc}$ are the Gibbs energies of the respective pure elements Pr and Ni in the non-magnetic dhcp and fcc structures respectively. The parameters a and b are evaluated in the present work. ${}^{mg}G_m^{Ni_APr_B}$ is the magnetic contribution to the Gibbs energy.

As can be seen in Table 1, the Curie temperatures of the intermetallic compounds in the Ni-Pr system, available in the literature, are much below 298.15 K. For this reason, we have not taken into account the contribution of the magnetic term in our calculations.

3. Bibliographic description of the system

3.1. Equilibrium diagram

The Ni-Pr system was first studied by Vogel and Fulling [22] using Pr with 98 to 99 wt.% purity. No information on Ni purity was given. Six compounds were reported by these authors: $NiPr_3$, NiPr, Ni_2Pr, Ni_3Pr, Ni_4Pr and Ni_5Pr, but the existence of Ni_3Pr and Ni_4Pr was signaled to be not certain. These authors established the Ni-Pr phase diagram based on data obtained from the differential thermal analysis measurements realized on only nine alloys, which were selected at critical compositions on the basis of the similarity of Ni-Pr to the Ni-La and Ni-Ce systems.

Later, Lemaire and Paccard [23] confirmed the existence of Ni_3Pr by means of X-ray diffraction (XRD) analysis. Using the same technique, Gschneidner [24] reported that the true composition of Ni_4Pr should be Ni_7Pr_2. This formulation was confirmed by Taylor [26] and by Buschow and Van Der Goot [25] who suggested that Ni_7Pr_2 occurs two crystallographic forms: the hexagonal high-temperature form and the rhombohedral low-temperature form. However, the transformation temperature was not determined by these authors.

Several XRD studies confirmed the existence of Ni_3Pr [27], NiPr [28,29], Ni_2Pr [26] and Ni_5Pr [30]. The occurrence of Ni_3Pr_7 was ascertained by Kissel et al. [31] and Olcese [32].

Vogel and Fulling [22] suggested that Ni_2Pr decomposes peritectically at 880 °C, but Wernick and Geller [33] did not observe any phase transition between room temperature and 880 °C in their DTA studies. Qi et al. [34] determined the liquidus in equilibrium with the Ni_5Pr compound in a temperature range of 1273 K to 1513 K.

Pan and Cheng [35] reinvestigated the Ni-Pr system and confirmed all previously reported compounds. The constructed phase diagram is similar to the evaluation given by Gschneidner [24], but the temperatures and compositions of invariant reactions are rather different.

The Ni-Pr phase diagram was reviewed by Pan and Nash [36] based on the differential thermal analysis (DTA) and XRD data of Vogel and Fulling [22] and Pan and Cheng [35]. Seven intermetallic compounds were confirmed: $PrNi_2$, $PrNi_3$ and Pr_2Ni_7 form by peri-

Table 2
Ni-Pr crystal structures data.

Phase	Composition (at.% Pr)	Pearson symbol	Prototype	Symbol used in Thermo-Calc data file	Reference
(Ni)	0.0	cF4	Cu	FCC_A1	[43]
Ni_5Pr	16.7	hP6	$CaCu_5$	Ni_5Pr	[43]
Ni_7Pr_2	22.2	hP36	Ni_7Ce_2[a] Ni_7Gd_2[b]	Ni_7Pr_2	[25]
Ni_3Pr	25.0	hR24	$PuNi_3$	Ni_3Pr	[27]
Ni_2Pr	33.3	cF24	Cu_2Mg	Ni_2Pr	[22]
NiPr	50.0	oC8	CrB	NiPr	[28]
Ni_3Pr_7	70.0	hP20	Fe_3Th_7	Ni_3Pr_7	[32]
$NiPr_3$	75.0	oP16	Fe_3C	$NiPr_3$	[23]
(βPr)[c]	100	cI2	W	BCC_A2	[44]
(αPr)[d]	100	hP4	αLa	DHCP	[44]

[a] High-temperature form.
[b] Low-temperature form.
[c] From 931 to 745 °C.
[d] From 745 °C.

Table 3
Calculated thermodynamic parameters for the Ni-Pr system.

Phase	Thermodynamic parameters
Liquid	${}^0L_{Ni,Pr} = -126,268 + 22.472 \times T$
	${}^1L_{Ni,Pr} = -82,377 + 52.61 \times T$
FCC_A1	${}^0L_{Ni,Pr} = 10^5$
DHCP	${}^0L_{Ni,Pr} = 10^5$
BCC_A2	${}^0L_{Ni,Pr} = 10^5$
Ni_5Pr	$G_{Ni_5Pr} = 0.83{}^0G_{Ni}^{fcc} + 0.17{}^0G_{Pr}^{dhcp} - 27,436 + 5.557 \times T$
Ni_7Pr_2	$G_{Ni_7Pr_2} = 0.78{}^0G_{Ni}^{fcc} + 0.22{}^0G_{Pr}^{dhcp} - 29,083 + 4.729 \times T$
Ni_3Pr	$G_{Ni_3Pr} = 0.75{}^0G_{Ni}^{fcc} + 0.25{}^0G_{Pr}^{dhcp} - 29,457 + 4.212 \times T$
Ni_2Pr	$G_{Ni_2Pr} = 0.67{}^0G_{Ni}^{fcc} + 0.33{}^0G_{Pr}^{dhcp} - 30,295 + 2.982 \times T$
NiPr	$G_{NiPr} = 0.50{}^0G_{Ni}^{fcc} + 0.50{}^0G_{Pr}^{dhcp} - 29,261 + 1.304 \times T$
Ni_3Pr_7	$G_{Ni_3Pr_7} = 0.30{}^0G_{Ni}^{fcc} + 0.70{}^0G_{Pr}^{dhcp} - 19,350 + 0.099 \times T$
$NiPr_3$	$G_{NiPr_3} = 0.25{}^0G_{Pr}^{fcc} + 0.75{}^0G_{Pr}^{dhcp} - 16,646 + 0.308 \times T$

tectic reactions and Pr_3Ni, Pr_7Ni_3, PrNi and $PrNi_5$ are congruently melting compounds. Five eutectic reactions occur: L ↔ α-Pr + Pr_3Ni (at 773 K and ~19 at.% Ni), L ↔ Pr_3Ni + Pr_7Ni_3 (at 763 K and ~27 at.% Ni), L ↔ Pr_7Ni_3 + PrNi (at 783 K and ~32.5 at.% Ni), L ↔ NiPr + $PrNi_2$ (at 953 K and ~53 at.% Ni), L ↔ PrNi5 + Ni (at 1553 K and ~89 at.% Ni).

More recently, Huang et al. [37] revised the Ni-Pr phase diagram using DTA, differential scanning calorimetry (DSC), and scanning electron microscopy (SEM). The results were compatible with those mentioned in the Pan and Cheng [35] diagram to the extent that the diagram contains seven intermediate phases, five eutectic reactions, and four peritectic reactions. However, in the Pr-rich region the invariant temperatures are from 30 to 90 °C higher than those found in previous studies [22,35]. We note that in the study by Huang et al. [37], the invariant temperatures were determined from the onset temperatures of peaks during heating in the DTA or DSC measurements. On the other hand, the temperatures determined by Vogel and Fulling [22] were extracted from DTA cooling curves, which can explain the difference between the published invariant temperatures in the Pr-rich part of the phase diagram.

3.2. Thermodynamic data

The enthalpy of formation of Ni_2Pr measured by Deodhar and Ficalora [38] is −8400 (J/mol at). Pasturel et al. measured by calorimetric method the enthalpy of formation of Ni_5Pr [39] and reported a value of −26,800 (J/mol at). Using direct reaction calorimetry Guo and Kleppa [40,41] measured the standard enthalpies of formation of NiPr and Ni_5Pr. The values obtained were −28,100 ± 1100 and −25,600 ± 1100 (J/mol at) respectively.

Nikolaenko [42] measured the enthalpy of mixing of liquid at 1750 K at the composition of 42 at% Pr. The reported value was −30,700 ± 1600 J/mol.

3.3. Crystallographic data

The Pr–Ni crystal structures are summarized in Table 2.

Table 4
Calculated and experimental invariant equilibria in the Ni-Pr system.

Reaction	Type	Temperature (K)		at % Pr in liquid	
		This work	Literature data [Ref.]	This work	Literature data [Ref.]
Liquid ↔ Ni_5Pr + fcc (Ni)	Eutectic	1553	1553 [35,48] 1543 [22] 1555 [47]	8.3	11.0 [35] 9.0 [22] 8.4 [48] 9.2 [47]
Liquid ↔ Ni_5Pr	Congruent	1658	1658 [35,48] 1638 [22] 1657 [47]	–	–
Liquid + Ni_5Pr ↔ Ni_7Pr_2	Peritectic	1435	1433 [35] 1373 [22] 1434 [47] 1435 [48]	30	30.0 [35] 29.9 [47] 28.7 [48]
Liquid + Ni_7Pr_2 ↔ Ni_3Pr	Peritectic	1302	1283 [35,47] 1245 [22] 1302 [48]	34.4	35.0 [35] 33.1 [47] 33.4 [48]
Liquid + Ni_3Pr ↔ Ni_2Pr	Peritectic	1194	1183 [35,47] 1154 [22] 1194 [48]	37.4	38.0 [35] 39.2 [47] 36.6 [48]
Liquid ↔ Ni_2Pr + NiPr	Eutectic	1006	1005 [37] 953 [35] 978 [22] 1006 [48] 952 [47]	44.8	47.0 [22,35] 45.0 [37] 41.9 [47] 44.3 [48]
Liquid ↔ NiPr	Congruent	1039	1039 [37,48] 1003 [22,35,47]	–	–
Liquid ↔ NiPr + Ni_3Pr_7	Eutectic	822	823 [37,48] 783 [35,47]	64.9	65.0 [37] 67.0 [35] 65.9 [48] 67.5 [47]
Liquid ↔ Ni_3Pr_7	Congruent	839	839 [37,48] 808 [35] 807 [47]	–	–
Liquid ↔ Ni_3Pr_7 + $NiPr_3$	Eutectic	833	830 [37] 763 [35] 777 [47] 832 [48]	73.1	72.0 [37] 73.0 [35] 72.8 [48] 72.9 [47]
Liquid ↔ $NiPr_3$	Congruent	835	835 [37,48] 798 [35] 803 [22] 797 [47]	–	–
Liquid ↔ $NiPr_3$ + dhcp (Pr)	Eutectic	827	827 [37,48] 733 [35] 753 [22] 738 [47]	79.2	77.0 [37] 80.0 [35] 85.0 [22] 78.8 [48] 78.5 [47]

3.4. Previous assessment

The Ni-Pr system has already been assessed by Du et al. [45] and by Huang and Lograsso [46]. Some difficulties are noticed: the calculated invariant temperatures by Du et al. [45] differ from the experimental ones. Whereas Huang and Lograsso [46] used the less exothermic enthalpy of formation determined by Deodhar and Ficalora [38] for the NiPr compound and ignored the standard enthalpies of formation measured by Guo and Kleppa [40,41] for the NiPr and Ni_5Pr compounds. Moreover, according to the results of Huang and Lograsso [46], the most exothermic enthalpy of formation occurs at the equiatomic composition (50 at.% Pr) in contrast with the available experimental Ni–RE measurements who situated this extremum at 25 at.% RE (Fig. 5).

4. Assessment procedure

On the basis of lattice stabilities cited from Dinsdale [16], the optimization of the model parameters was conducted using the PARROT module [49] in the Thermo-Calc software package developed by Sundman et al. [15]. This module works by minimizing the square sum of the differences between experimental data and calculated values. In the optimization procedure, each set of experimental data was given a certain weight. The weights were changed systematically during the optimization until most of experimental data was accounted for within the claimed uncertainty limits. The optimization was carried out by steps. The parameters of the liquid phase were first optimized by the experimental data of liquidus boundaries. The congruent intermetallic compounds were investigated next. The parameters of the intermetallic compounds were consequently optimized by using phase diagram data and available thermodynamic information. Then experimental data related to invariant equilibria have been added to the calculation. All the parameters were finally evaluated together to give the best description of the system. We note that we have neglected the solubilities in the FCC_ A1, BCC_A2 and DHCP terminal solid solutions in accord with the available experimental results. This was realized by assigning a large positive value to the ${}^{j}L^{\phi}_{\mathrm{Ni,Pr}}$ interaction parameter of Eq. (5), i.e.

$$ {}^{0}L^{\mathrm{DHCP}}_{\mathrm{Ni,Pr}} = {}^{0}L^{\mathrm{BCC_A2}}_{\mathrm{Ni,Pr}} = {}^{0}L^{\mathrm{FCC_A1}}_{\mathrm{Ni,Pr}} = 10^5 \quad (7) $$

The thermodynamic parameters obtained in the present work for all condensed phases in the Ni-Pr binary system are summarized in Table 3. In Table 4, are listed all invariant reactions for this binary system. Good agreement has been realized between the calculated results and the available experimental data.

5. Results and discussion

5.1. The assessed phase diagram

The calculated Ni-Pr phase diagram is illustrated in Fig. 1. In Fig. 2 we compare our calculated phase diagram with the experimental results reported by Huang et al. [37] and the previous ones published by Pan and Cheng [35].

The detailed phase relationships for Pr mole fractions from 0.6 to 0.8 and temperatures from 780 to 860 K are further shown together with experimental data in Fig. 3. The phase diagram obtained in this work is in good agreement with that established by Pan and Cheng [35] and the one reevaluated by Huang et al. [37]. Our calculated phase diagram corrects the unusual asymmetric curves of the liquidus phase [49,50] around NiPr and Ni_5Pr.

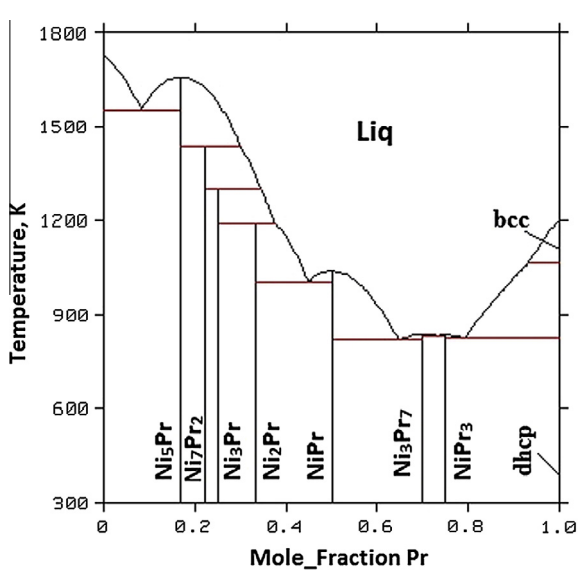

Fig. 1. The Ni-Pr phase diagram calculated in this work.

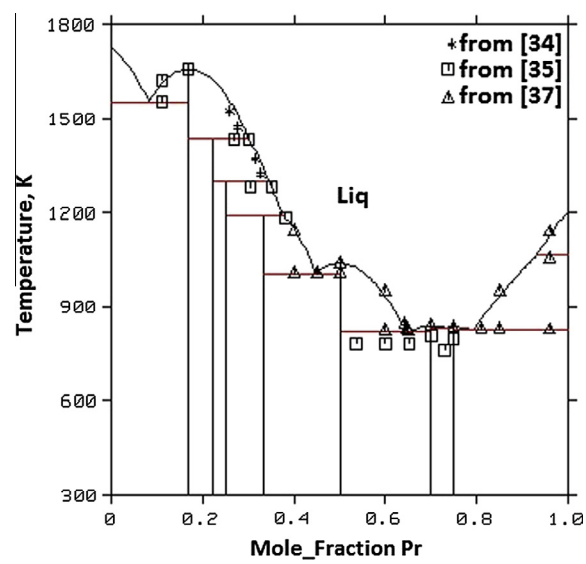

Fig. 2. Calculated Ni-Pr phase diagram compared with available experimental data.

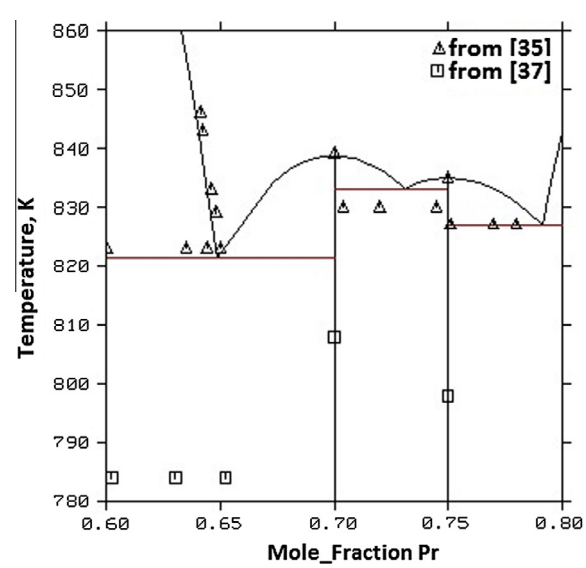

Fig. 3. Enlarged phase diagram of the Pr-rich part with experimental data.

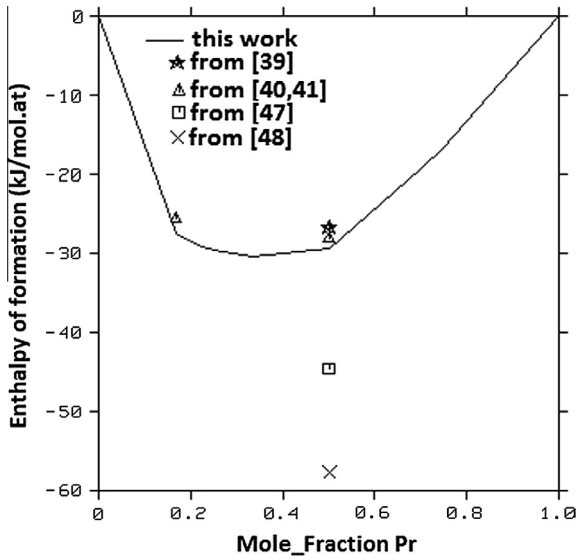

Fig. 4. Calculated enthalpies of formation of the Ni-Pr intermetallic compounds with available literature data.

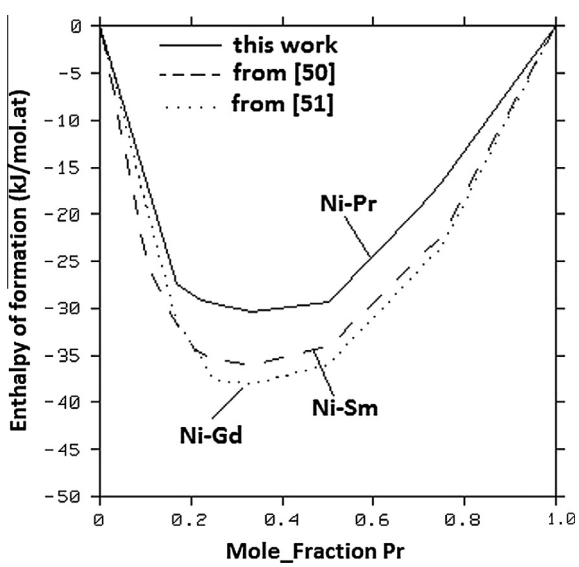

Fig. 5. Calculated enthalpies of formation of the Ni-Pr intermetallic compounds together with those of others Ni–RE systems.

5.2. Enthalpies of formation for intermetallic compounds

As mentioned previously, the enthalpies of formation of intermetallic compounds are measured by several authors. In general, calorimetric direct measurements are considered to be the most reliable, so the values obtained by Guo and Kleppa [40,41] were used in the present study with high weights. The calculated enthalpies of formation of Ni-Pr intermediate phases are represented together with the literature data in Fig. 4. For comparison, the enthalpies of formation calculated in this work with the enthalpies of others Ni–RE systems (Ni–Sm [47] and Ni–Gd [48]) were plotted in Fig. 5. As can be seen, the enthalpy of formation of Ni_2Pr is the most negative, which is in accordance with the results obtained for others Ni–RE systems.

We note that the enthalpy of formation of NiPr compound (Fig. 4) calculated by Huang and Lograsso [46] and by Du et al. [45] is the most negative and differs from the experimental values by 16.5 and 29.6 (kJ/mol at) respectively. This situation can be explained by the elevated value of entropy of formation accepted by these authors for the NiPr phase [45,46].

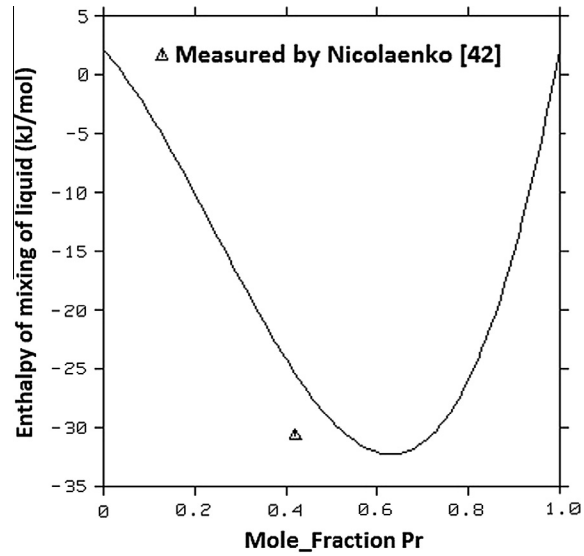

Fig. 6. Calculated enthalpy of mixing of the liquid phase at 1750 K. Experimental datum is the calorimetric result of Ref. [42].

The enthalpy of mixing of liquid of Ni-Pr system calculated in this work over the whole range of composition and at 1750 K is presented in Fig. 6 with the calorimetric result reported by Nikolaenko [42] at 1750 K for a composition of 42% Pr. The discrepancy between the two results at this composition is acceptable.

6. Conclusion

The phase relationships and thermodynamic description in the Ni-Pr system were critically evaluated from the experimental information available in the literature. A self-consistent set of thermodynamic parameters was derived. The calculated phase diagram agrees well with literature data and reproduce all the phase relations in this system. In addition, we have corrected the unusual asymmetric liquidus around the NiPr and Ni_5Pr phases to be thermodynamically accepted.

References

[1] K.H.J. Buschow, Rep. Prog. Phys. 40 (1977) 1179.
[2] D. Chandra, J.J. Reilly, R. Chellappa, JOM 58 (2006) 26.
[3] K.H.J. Buschow, Mater. Res. Bull. 19 (1984) 935.
[4] S. Sugiomoto, J. Phys. D Appl. Phys. 44 (2011) 11.
[5] A. Inoue, Prog. Mater. Sci. 43 (1998) 365.
[6] L. Battezzati, S. Pozzovivo, P. Rizzi, Mater. Trans. 43 (2002) 2593.
[7] M. Zinkevich, F. Aldinger, in: Euromat 2003, Symp. T2.
[8] T. Sakai, M. Matsuoka, C. Iwakura, in: K.A. Gschneidner Jr., L. Eyring (Eds.), Handbook on the Physics and Chemistry of Rare Earths, vol. 21, North{C}–Holland, Elsevier Science Publishers B.V, The Netherlands, 1995, p. 133.
[9] C. Zimm, A. Jastrab, A. Sternberg, V.K. Pecharsky Jr., K. Gschneidner, M. Osborne, et al., Adv. Cryog. Eng. (1998) 1759.
[10] C. Zimm, A. Boeder, J. Chell, A. Sternberg, A. Fujita, S. Fujieda, et al., Int. J. Refrig. 29 (2006) 1302.
[11] O. Tegus, E. Brück, K.H.J. Buschow, F.R. de Boer, Nature 415 (2002) 150.
[12] M. Balli, O. Sari, C. Mahmed, Ch. Besson, Ph. Bonhote, D. Duc, J. Forchelet, Appl. Energy 98 (2012) 556.
[13] M. Baricco, M. Palumbo, J. Met. Nano. Mater. 20–21 (2004) 415.
[14] L. Kaufman, H. Bernstein, Computer Calculation of Phase Diagrams, Academic Press, New York, 1970.
[15] B. Sundman, B. Jansson, J.O. Andersson, CALPHAD 9 (1985) 153.
[16] A.T. Dinsdale, CALPHAD 15 (1991) 317.
[17] O. Redlich, A.T. Kister, Ind. Eng. Chem. 40 (1948) 345.
[18] R.E. Walline, W.E. Wallace, J. Chem. Phys. 41 (6) (1964) 1587.
[19] R. Lemaire, D. Paccard, R. Pauthenet, C.R. Acad, Sci. Paris 265B (1967) 1280.
[20] J. Sebek, R.M. Mueller, J. Kuriplach, J. Phys. 46 (1996) 2215.

[21] D. Paccard, R. Pauthnet, C.R. Acad, Sci. Paris 264B (1967) 1056.
[22] R. Vogel, W. Fulling, Z. Metallkd. 38 (1947) 97.
[23] R. Lemaire, D. Paccard, Bull. Soc. Fr. Min. Crystallogr. 90 (1967) 311.
[24] K.A. Gschneidner, Rare Earth Alloys (1961) 221.
[25] K.H.J. Buschow, A.S. Van Der Goot, J. Less-Common Met. 22 (4) (1970) 419.
[26] K.N.R. Taylor, Adv. Phys. 20 (1971) 551.
[27] D. Paccard, R. Pauthenet, C.R. Acad, Sci. Ser. A B 264B (1967) 1056.
[28] A.E. Dwight, R.A. Conner Jr., J.W. Downey, Acta Crystallogr. 18 (1965) 835.
[29] S.C. Abrahams, J.L. Bernstein, R.C. Sherwood, J.H. Wernick, H.J. Williams, J. Phys. Chem. Solids 25 (1964) 1069.
[30] A.E. Dwight, Trans. Asm. Soc. Met. 53 (1961) 479.
[31] F. Kissell, T. Tsuchida, W.E. Wallace, J. Chem. Phys. 44 (1966) 4651.
[32] G.L. Olcese, J. Less-Common. Met. 33 (1973) 71.
[33] J.H. Wernick, S. Geller, Trans. Metall. Soc. AIME 218 (1960) 866.
[34] G. Qi, Z. Li, K. Itagaki, Mater. Trans. JIM 30 (1989) 583.
[35] Y.Y. Pan, C.S. Cheng, Chinese National Symposium on Phase Diagrams, Kumming, 1984, p. 21.
[36] Y.Y. Pan, P. Nash, Bull. Alloy Phase diag. 10 (1989) 253.
[37] M. Huang, D. Wu, K.W. Dennis, J.W. Anderegg, R.W. McCallum, T.A. Lograsso, J. Phase Equilibr. Diff. 26 (2005) 209.
[38] S.S. Deodhar, P.J. Ficalora, Metall. Trans. 6A (1975) 1909.
[39] A. Pasturel, C. Colinet, C. Allibert, P. Hicter, A. Percheron-Guegan, J.C. Achard, Phys. Status Solidi 125b (1984) 101.
[40] Q. Guo, O.J. Kleppa, J. Alloys Comp. 270 (1998) 212.
[41] Q. Guo, O.J. Kleppa, Met. Mater. Trans. 25B (1994) 73.
[42] I.V. Nikolaenko, J. Alloys Comp. 225 (1995) 474.
[43] J.H. Wernick, S. Geller, Acta Crystallogr. 12 (1959) 662.
[44] K.A. Gschneidner, F.W. Calderwood, Handbook Phys. Chem. Rare Earth 8 (1986) 1.
[45] Z. Du, D. Wang, W. Zhang, J. Alloys Comp. 284 (1999) 206.
[46] M. Huang, T.A. Lograsso, J. Alloys Comp. 395 (2005) 75.
[47] G. Borzone, N. Parodi, R. Raggio, R. Ferro, J. Alloys Comp. 317–318 (2001) 532.
[48] C. Colinet, A. Pasturel, K.H.I. Buschow, Met. Trans. 17A (1986) 777.
[49] H. Okamoto, T.B. Massalski, J. Phase Equilibr. 14 (3) (1993) 316.
[50] H. Okamoto, T.B. Massalski, J. Phase Equilibr. 15 (5) (1994) 500.